

EDUCATION

Oregon State University

Expected Graduation: Spring 2026

Bachelor of Science, Biology, Minor: Biochemistry/Molec. Bio. (GPA 3.4)

Corvallis, OR

- **Relevant Coursework:** General Chemistry, Organic Chemistry, General Biochemistry, Genetics, Evolution, Vertebrate Physiology, Cell & Molecular Biology, Molec. Genetics/Genomics Lab., Microbiology with Lab, Calculus (Differential & Integral), Statistics, Python for Molecular Biologists, Computer Programming for non-CS Majors, Intro. Biological Data Science, Applied Bioinformatics, Adv. Computing for Bio. Data Analysis (ML).

EXPERIENCE

Oregon State University

Corvallis, OR

Bioinformatics Research Assistant, OSU - Chang Lab

Jan. 2025 – Present

- Developed and managed large-scale genome assembly datasets for *Phytophthora* isolates (100+ isolates; 300+ assemblies across primary, hap1, hap2) enabling comparative genomics analyses
- Designed and executed scalable genomics pipelines using **GENESPACE**, **nf-core/pairgenomealign**, and **MUMmer** to analyze synteny, structural variation, and genome evolution
- Built and optimized **HPC workflows (SLURM array jobs)** to automate genome alignment and annotation across dozens of comparisons, improving efficiency and reproducibility
- Integrated genome annotation tools (**Helixer**, **AGAT**) within **Singularity containers** to create portable and reproducible analysis pipelines
- Generated publication-quality visualizations in **R** (phylogenies, synteny dotplots, repeat-density plots) for presentations and research outputs
- Collaborated with postdoctoral researchers to troubleshoot workflows and refine large-scale comparative genomics analyses

Microbiology Research Assistant, OSU - Halsey Lab

Mar. 2026 – Present

- Lead molecular identification of bacterial members in a biofuel-relevant microbial consortium (*Bacillus pumilus* and contaminants) using **PCR-based approaches (16S rRNA amplification)**
- Designed and implemented full molecular workflows including **DNA extraction, PCR optimization, gel electrophoresis, and sequencing preparation**
- Trained lab members in PCR methodology, troubleshooting amplification issues, and improving experimental reliability
- Developed organism identification strategies using **universal and custom primers**, with downstream sequence-based analysis for taxonomic characterization
- Cultured/maintained microbial isolates (bacterial, algal) using sterile technique and controlled growth conditions
- Connected wet-lab results with computational analysis to investigate microbial composition and lipid production potential for **biofuel applications**

LEADERSHIP

Club Secretary, Vice-President

Jun. 2024 – Jun. 2026

Oregon State University Integrative Biology Club

Corvallis, OR

- Lead operations and programming for a 500+ member student organization, collaborating with faculty and peers to expand academic, professional, and community development opportunities.
- Organized and facilitated professional workshops and graduate/undergraduate research information sessions connecting students with faculty and helping them build confidence and experience.

Biology Learning Assistant (LA)

Sep. 2024 – Jun. 2025

Oregon State University, College of Science

Corvallis, OR

- Supported the BI 22X introductory biology series by facilitating active learning, assisting in lectures, and holding weekly office hours for 1000+ students.
- Attended weekly content prep meetings to discuss content and delivery with other LAs and instructors.
- Delivered individualized tutoring and small-group support to improve student comprehension and retention of core biology concepts.

Alumni Relations Chairman

Apr. 2025 – Feb. 2025

Phi Kappa Psi Fraternity, Oregon Beta Chapter

Corvallis, OR

- Managed communications with 1,000+ alumni through bi-monthly newsletters and event updates.
- Coordinated major events, including a flagship Alumni Golf Mixer (budget \$1,500; 50+ attendees) and the annual Founder's Day Banquet (budget \$20,000; 300+ attendees).